# PRDA – 05 Customer Data Analysis

**Internship Capstone Project**

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Tool Used: MYSQL Workbench, Power BI

Year: 2026

## Project Introduction & Description

This project focuses on analyzing customer shopping behavior using transactional data collected from different shopping malls. The analysis helps to understand customer demographics, purchasing patterns, payment methods, and revenue generation.

SQL is used to extract and analyze the data, and the insights obtained from this project support better business decisions and customer-focused strategies.

## *Dataset Description*

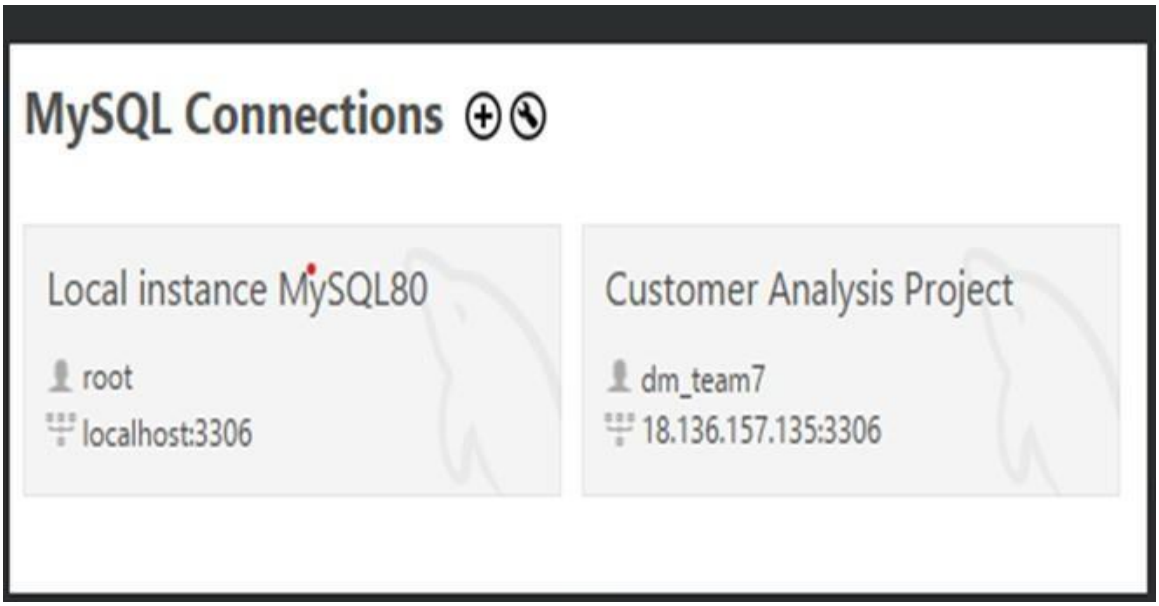
The dataset used in this project contains customer shopping transaction data collected from multiple shopping malls. It includes customer demographic details and purchase-related information, which were used for analysis and visualization.

# Dataset Attributes Table

| Column Name | Description |
|---|---|
| Invoice no | Unique invoice number |
| Customer id | Unique customer ID |
| Gender | Gender of the customer |
| Age | Age of the customer |
| Category | Product Category |
| Quantity | Number of items purchased |
| Price | Price per unit |
| Payment Method | Payment method used |
| Invoice date | Date of transaction |
| Shopping mall | Shopping mall name |

# DATABASE SETUP & CONNECTION

The dataset was accessed from a MYSQL database using MySQL workbench. A Successful connection to the remote database was established, and the required database and customer table were verified for analysis.




# Power BI Analysis

## Introduction

In this project, Power BI is used to analyze customer shopping data.

The purpose of using Power BI is to create visual reports and understand customer behavior easily through charts and graphs.

## Dataset Description

The dataset contains customer shopping details such as

- Invoice number
- Customer Id
- Gender
- Age
- Product Category
- Quantity
- Price
- Payment Method

This data helps us to explain shopping patterns of customers.

## Tools Used in Power BI

- Microsoft Power BI Desktop (visualizations)
- Microsoft Excel (Dataset)

## Data Preparation in Power BI

Before creating the chart, the following steps are done.

1. Opened Power BI Desktop
2. Click on Get data.
3. Selected Excel Workbook.
4. Choose the downloaded dataset file.
5. Clicked load to import the data.
6. Checked all the columns were loaded correctly.
7. Verified there was no missing values in Gender Column.

## Task 1

- How is the shopping distribution according to gender?

## Objective

The objective of this analysis is to understand how shopping transactions are distributed between Male and Female customers.

## Description

A Donut Chart is used in Power BI to visualize the distribution of shopping transactions based on gender. This type of chart helps to clearly show the percentage contribution of each gender to the total number of transactions.

## Steps to Follow

- Import the dataset into Power BI.
- Select the Donut Chart from the Visualizations pane.
- Drag the Gender field to the Legend section.
- Drag the Transaction Count / Customer ID to the Values section.

- Add the chart title “Shopping Distribution According to Gender.”
- Enable Data Labels to display the values and percentages.

## Visualization

## Insights

From the chart, it is observed that Female customers contribute 62% of the total transactions, while Male customers contribute 38%. This indicates that Female customers make more purchases compared to Male customers.

# Task 2

Which gender did we sell more products to?

**SQL Query :**

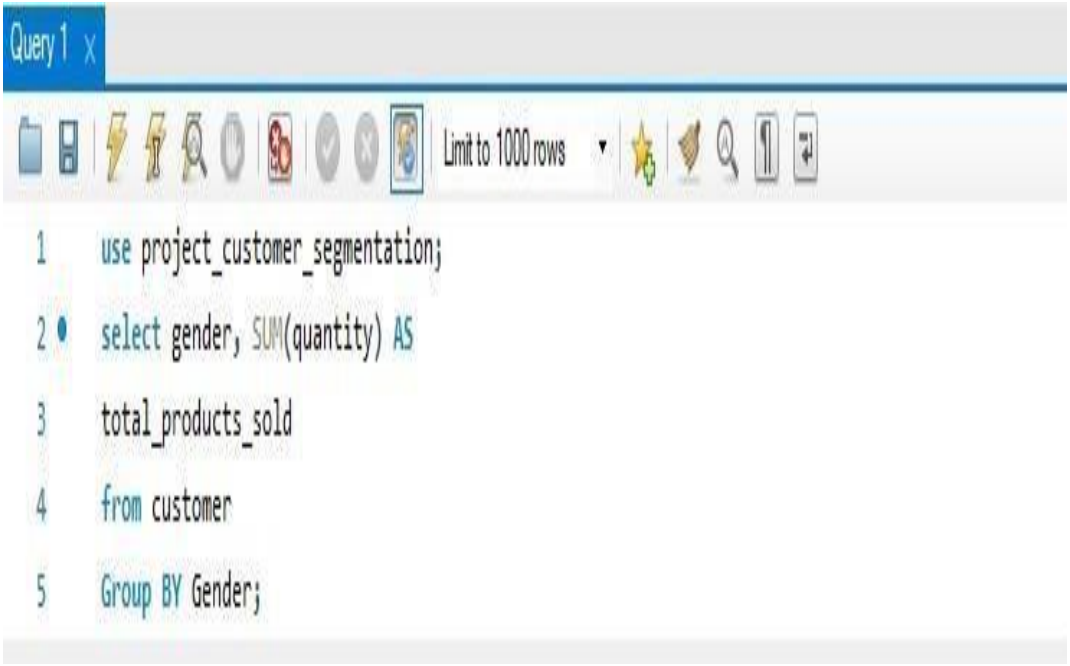
SELECT gender,

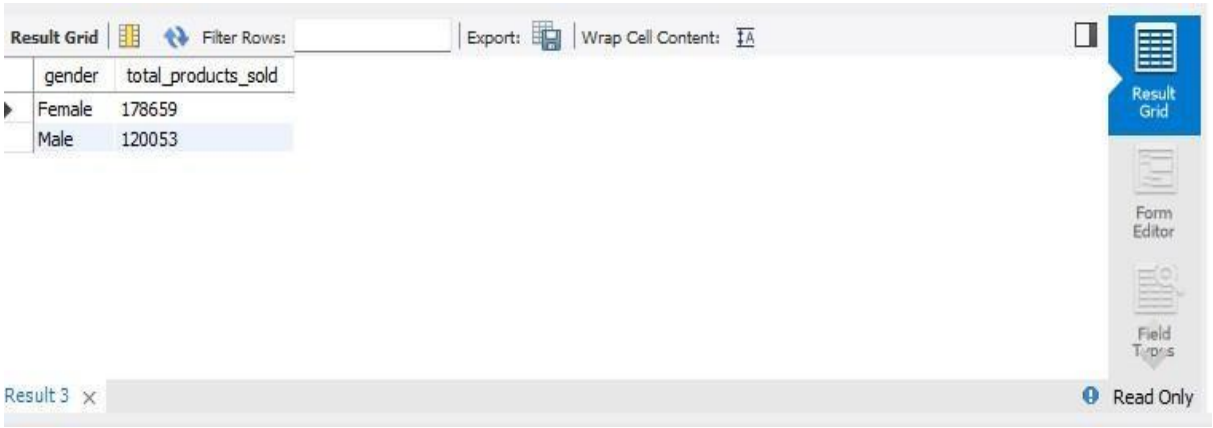
SUM(quantity) AS

Total_products_sold from

customer

Group BY gender;






| gender | total_products_sold |
|---|---|
| Female | 178659 |
| Male | 120053 |

## Result

Based on the total quantity sold, female customers purchased more products than male customers.

## Task 3

Which gender generated more revenue?

```
Select gender,
Sum(quantity * price) As total_revenue
from customer
Group By gender;
```

Query 1

Limit to 1000 rows

```
select
gender,
sum(Quantity * price) As total_revenue
from customer
Group BY gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| gender | total_revenue |
|---|---|
| Female | 150207136.02 |
| Male | 101298658.23 |

Result Grid

Form Editor

Field Types

Result 5

Read Only

## Result

From the result, it is observed that female customers generated more revenue than male customers.

## Task 4 & 8

- Distribution of Purchase Categories Relative to Gender?

To analyze how product categories are purchased relative to gender, two visualizations were created in Power BI.

## Chart 1

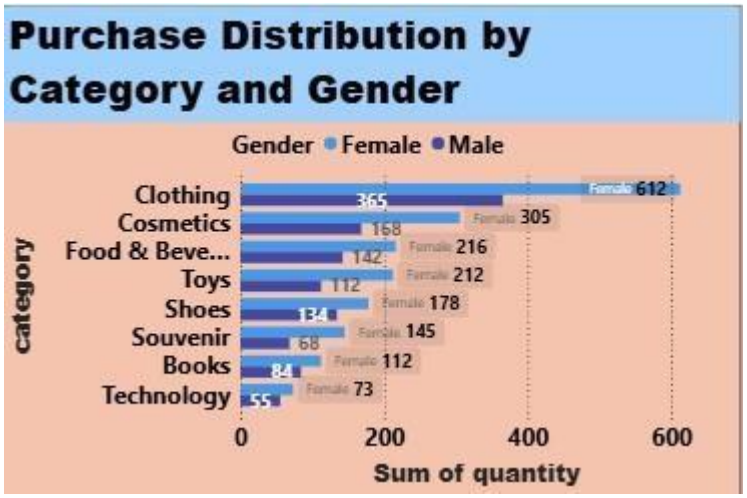
Total Quantity Purchased by Category and Gender (Clustered Column Chart)

## Steps to Follow

- Import the dataset into Power BI.
- Select the Clustered Bar Chart from the visualization pane.
- Drag the Category field to the Y-axis.

- Drag Quantity to the X-axis and set it to Sum of Quantity.
- Drag Gender to the Legend field to compare Male and Female purchases.
- Add data labels and provide a suitable chart title.

## Visualization

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## Description

This chart shows the total quantity of products purchased in each category and compares the purchases made by male and female customers.

## Observation

Clothing has the highest quantity purchased among all categories. Female customers tend to purchase more

items in categories like Clothing, Cosmetics, and Toys compared to male customers.

**Insights**

The chart highlights differences in purchasing patterns between male and female customers and helps identify which product categories are more popular among each gender.

**Chart 2**

Total Quantity Purchased by Category and Gender (Clustered Column Chart)

**Steps to Follow**

- Import the dataset into Power BI Desktop.
- From the Visualizations pane, select the Clustered Column Chart.
- Drag the Category field to the X-axis.
- Drag the Quantity field to the Y-axis and set it as Sum of Quantity.
- Drag the Gender field to the Legend section to compare Male and Female purchases.
- Enable Data Labels to clearly display the quantity values.

- Add a suitable title such as “Total Quantity Purchased by Category and Gender.”

This chart shows a vertical comparison of total quantity purchased across different categories by male and female customers.

## Visualization

## Observation

The chart shows the total quantity of products purchased in each category by male and female customers. Clothing and Cosmetics categories have higher purchase quantities compared to other categories.

## Insights

The visualization helps compare purchasing behaviour between male and female customers across different product categories and identifies which categories have higher purchase quantities.

## Task 5

- Shopping Distribution by Age Group

## Objective

To analyze how shopping expenditure is distributed across different age groups.

## Chart Used

- Clustered Column Chart

## Steps to Follow

- Open Power BI Desktop.
- Import the dataset.
- Select Clustered Column Chart from the Visualizations pane.
- Drag Age Group to the X-axis.
- Drag Price to the Y-axis.

- Set Price aggregation to Sum.
- Turn Data Labels ON for better visibility.
- Add the title “Shopping Distribution by Age Group.”

## Observation

The chart shows the total shopping value for each age group. The 56+ age group has the highest total spending, while the 18–25 age group has the lowest spending.

## Insights

- Customers in the 56+ age group spend the most on shopping.
- The 36–45 age group shows moderate spending.
- The 18–25 age group contributes the least to total shopping value.

Overall, shopping expenditure increases with age.

## Conclusion

The analysis indicates that older customers tend to spend more on shopping compared to younger customers.

## Task 6

- Which age category did we sell more products to?

## Step 1

Find which age group bought more products.

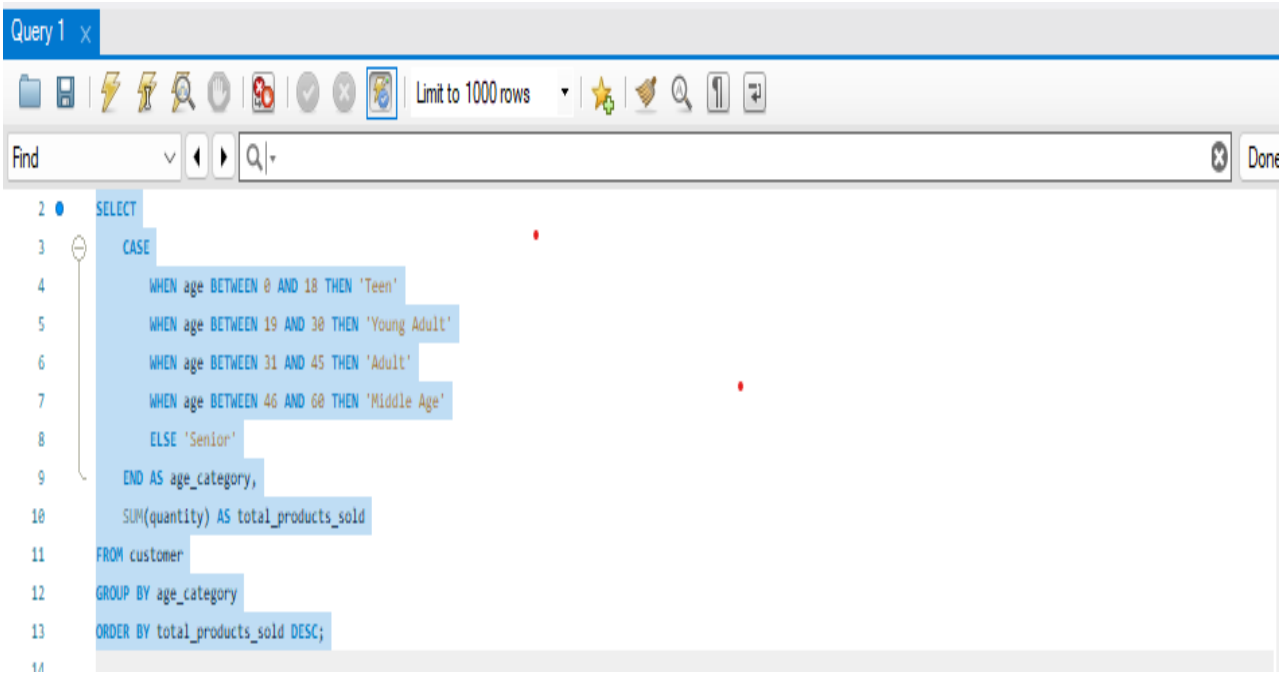
## Step 2

Group customers by age and calculate total quantity

## SQL Query

```
SELECT
    CASE
        WHEN age BETWEEN 0 AND 18 THEN 'Teen'
```

```
        WHEN age BETWEEN 19 AND 30 THEN 'Young Adult'
        WHEN age BETWEEN 31 AND 45 THEN 'Adult'
        WHEN age BETWEEN 46 AND 60 THEN 'Middle Age'
        ELSE 'Senior'
    END AS age_category,
    SUM(quantity) AS total_products_sold
FROM customer
GROUP BY age_category
ORDER BY total_products_sold DESC;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| age_category | total_products_sold |
|---|---|
| Adult | 86604 |
| Middle Age | 85399 |
| Young Adult | 69654 |
| Senior | 51537 |
| Teen | 5518 |

Result 2 × Read Only

## Result

Based on the analysis the adult age group shows the highest product purchases, making it the most active customer segment.

## Task 7

Which age Category generated more revenue?

## Step 1

Find which age group generated the highest revenue.

## Step 2

Calculate total revenue using quantity and price and group customers by age category.

**SQL Query**

```
SELECT
    CASE
        WHEN age BETWEEN 0 AND 18 THEN 'Teen'
        WHEN age BETWEEN 19 AND 30 THEN 'Young Adult'
        WHEN age BETWEEN 31 AND 45 THEN 'Adult'
        WHEN age BETWEEN 46 AND 60 THEN 'Middle Age'
        ELSE 'Senior'
    END AS age_category,
    SUM(quantity * price) AS total_revenue
FROM customer
GROUP BY age_category
ORDER BY total_revenue DESC;
```

```sql
SELECT
    CASE
        WHEN age BETWEEN 0 AND 18 THEN 'Teen'
        WHEN age BETWEEN 19 AND 30 THEN 'Young Adult'
        WHEN age BETWEEN 31 AND 45 THEN 'Adult'
        WHEN age BETWEEN 46 AND 60 THEN 'Middle Age'
        ELSE 'Senior'
    END AS age_category,
    SUM(quantity * price) AS total_revenue
FROM customer
GROUP BY age_category
ORDER BY total_revenue DESC;
```

# Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| age_category | total_revenue |
|---|---|
| Adult | 73134764.61 |
| Middle Age | 71804157.87 |
| Young Adult | 58649681.85 |
| Senior | 43519469.66 |
| Teen | 4397720.26 |

Result 3 × Read Only

# Task 9

Does the payment method have a relation with other columns?

## Objective

The objective of this analysis is to examine the total revenue generated through different payment methods and genders and to understand the shopping distribution across various age groups.

## Chart 1

- Total Revenue by Payment Method and Gender

## Steps to Follow

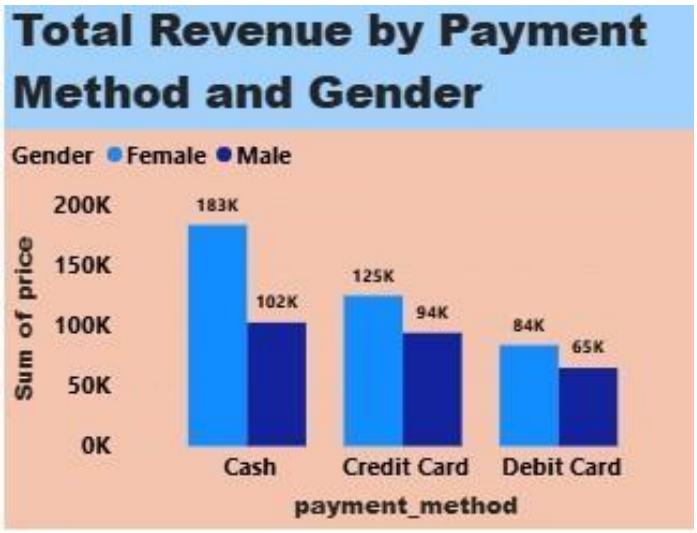
- Import the dataset into Power BI.
- Select Clustered Column Chart from the Visualizations panel.
- Drag payment method to the X-axis.
- Drag price to the Y-axis to calculate Sum of price (Total Revenue).
- Drag gender to the Legend section.
- Enable Data Labels for better visibility.
- Add the chart title “Total Revenue by Payment Method and Gender.”

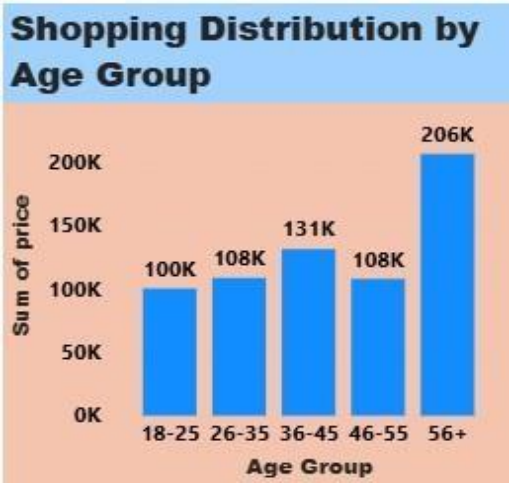
## Description

This chart shows the total revenue generated through different payment methods such as Cash, Credit Card, and Debit Card. The data is further categorized by gender to compare the revenue contribution of male and female customers.

## Insights

Female customers generate higher revenue compared to male customers across most payment methods. Cash is the most preferred payment method and contributes the highest revenue.



## Chart 2

- Shopping Distribution by Age Group

## Steps to Follow

- Insert another Clustered Column Chart.
- Drag Age Group to the X-axis.
- Drag price to the Y-axis to calculate Sum of price.
- Sort the Age Group in ascending order.
- Enable Data Labels to show the values clearly.
- Add the chart title “Shopping Distribution by Age Group.”

## Description

This chart illustrates the total shopping revenue generated by different age groups such as 18–25, 26–35, 36–45, 46–55, and 56+.

## Insights

The chart indicates that the 56+ age group contributes the highest shopping revenue, while the 18–25 age group contributes the lowest revenue.

## Task 10

How is the distribution of the payment method?

## Objective

To analyze how customers prefer to pay for their purchases and understand the proportion of different payment methods used in transactions.

## Chart Used

- Donut Chart

A donut chart is used because it clearly shows the percentage distribution of payment methods.

## Steps to Create in Power BI

- Open Power BI Desktop.
- Go to Visualizations panel.
- Select Donut Chart.
- Drag payment method → Legend.
- Drag invoice no → Values.
- Change aggregation to Count of invoice no.
- Turn Data Labels → ON.
- Turn Percentage → ON.
- Add chart Title: "Distribution of Payment Method".

## Visualizations

## Insights

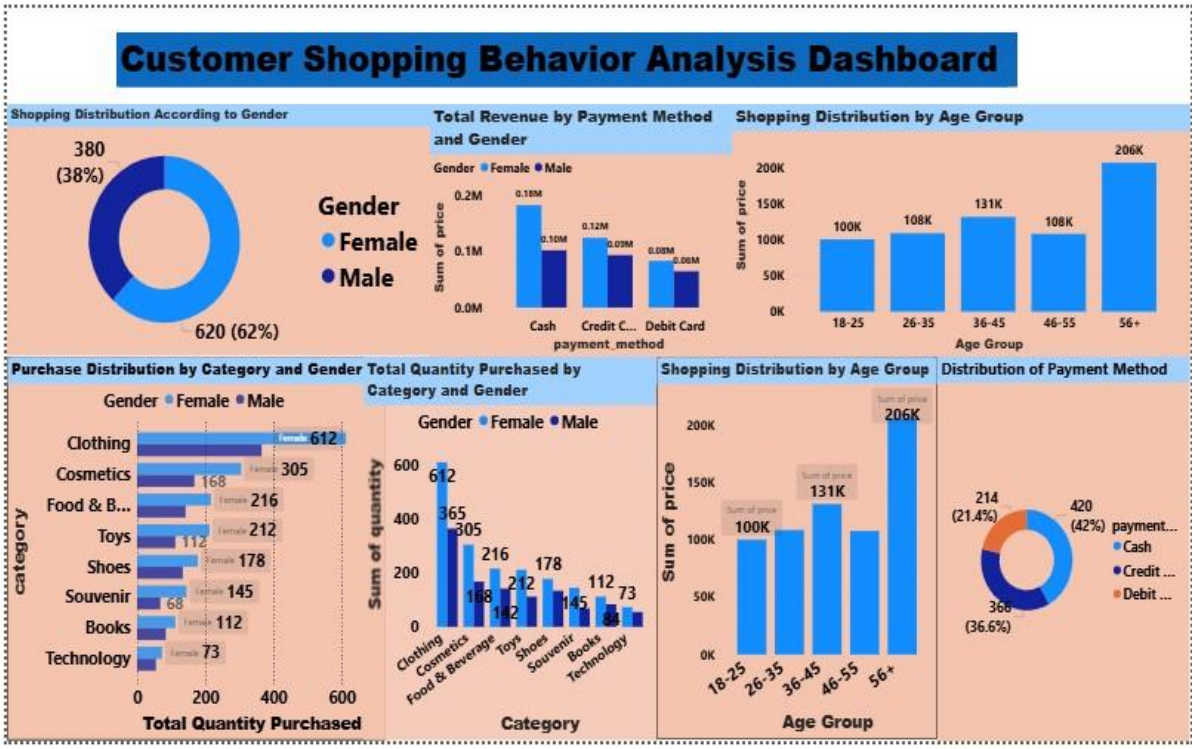
The chart shows the distribution of different payment methods used by customers. Cash payments represent the largest share of transactions at 42%, followed by Credit Card payments at 36.6%, while Debit Card payments account for 21.4%. This indicates that most customers prefer paying with cash compared to digital payment methods.

## Conclusion

Cash is the most commonly used payment method, whereas debit card transactions are relatively lower. Businesses may encourage digital payments to improve transaction efficiency and tracking.

## Task 11

Visualize the data using Tableau /Power BI and derive insights and give your inputs/suggestions to the company



## Conclusion

This project used Microsoft Power BI to analyze customer shopping behavior and visualize key insights from the dataset. The dashboard highlights patterns in purchasing based on gender, age group, product category, and payment methods. These visual insights help the company better understand customer preferences and improve business decision-making strategies. Overall, the dashboard provides an effective way to interpret the data and support data-driven decisions.